WHAT THE WELD'S NON-ASSOCIATIVITY IS *NOT*: A NEGATIVE TRIANGULATION ACROSS THREE OBSTRUCTION THEORIES, AND THE HANDEDNESS INVARIANT THAT SURVIVES

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Abstract. The worker/harness *weld* — the Zappa–Szép (orchestration) composition of directed containers, the program's central tool for *when* compositions exist — is non-associative in general. We ask which categorical invariant measures that non-associativity, and we answer by *negative triangulation*: three standard "obstruction homes" are probed by three *named* candidate maps. One *exists* (the weld is a regular bicomodule, hence a parametric right adjoint, after Spivak–Garner–Fairbanks); two are *refuted* — the renormalization idempotent $k = n\,;m$ of Di Lavore–Román–Székely is *not* a 2-cocycle (its receptacle does not exist: the cocycle *condition is* associativity) and *not* a Karoubi/Cauchy splitting obstruction (its receptacle is *vacuous*: k is already a split idempotent, and idempotent completion preserves associativity). Two independent sources confirm the Karoubi vacuity is generic, not weld-specific. The invariant that survives is neither cohomological nor splitting-theoretic: it is *one-sided absorptivity* (handedness) — k is left- but not right-absorptive. The cross-theme punchline is that two unrelated obstruction theories are both *orthogonal* to associativity for the *same* fact read twice. A dedicated literature sweep found no prior categorical home for handedness-of-a-single-map; we state this honestly as evidence, not as a theorem, and flag lax distributive laws by inequality as the one live fourth lead.

1. Setup: the weld and its non-associativity

Throughout, **Poly** is the category of polynomial functors on **Set** with the composition monoidal product $\triangleleft$; a comonoid for $\triangleleft$ is a *polynomial comonad*, equivalently a small category. For a set S we write $\Delta S = S\mathsf{y}^S$ (notation y for the representable) for the *store* comonad on S, the directed container of the full state space on S. All composites are written in diagrammatic order: $f\,;g$ means "f then g", and vertical composition of natural transformations is likewise diagrammatic.

Definition 1.1 (The weld)**.** Given two directed containers presented by store comonads ΔS (the *worker* state) and ΔT (the *harness* state), their *weld* is the orchestration composite

$$B \;=\; \Delta S \triangleleft \Delta T,$$

the Zappa–Szép product that interleaves worker and harness transitions. The weld is the program's canonical answer to "when does a composite of two stateful directed containers exist?": the Zappa–Szép datum is exactly the mutual-action data that makes the interleaving well defined.

The weld is *not* associative in general. This is not a defect of our encoding: it is intrinsic. The sharpest extant witness is probabilistic. Di Lavore, Román and Székely ("Normalized Probabilistic Semantics is Not Associative", arXiv:2510.01131) show that *normalization* of substochastic kernels is a *sesquilaw*, not a distributive law, and that the induced Kleisli-style composition of normalized kernels is a genuine *magmoid* — a category minus associativity.

Definition 1.2 (Sesquilaw and the renormalization idempotent, after arXiv:2510.01131)**.** A *sesquilaw* (S, T, m, n) consists of a distributive law $m\colon ST \Rightarrow TS$ together with an *almost-distributive* law

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$n\colon TS \Rightarrow ST$ satisfying all four distributive-law axioms *except* multiplicativity, which holds only up to the natural transformation

$$k \;=\; n\,;m \;:\; TS \Rightarrow TS \qquad \text{(the "renormalization" of Lemma 3.5).}$$

For normalization one takes $T = D$ (distribution) and $S = M$ (maybe), so $TS = DM$ and $ST = MD$; then $n = (-)^{\circ}\colon DM \Rightarrow MD$ is normalization and $m = (-)^{\bullet}\colon MD \Rightarrow DM$ is the subdistribution inclusion. The induced magmoid **Norm** of normalized kernels $X \to MDY$ is a category minus associativity; Proposition 4.9 of loc. cit. isolates the triples that do associate.

Remark 1.3. We treat $k = n\,;m$ as a faithful, finite-dimensional stand-in for the weld's associativity defect. The identification is justified computationally below (§5, reproducing arXiv:2510.01131's Prop. 4.9 purity cases exactly) and at the level of principle by the fact that both objects are instances of "an idempotent measuring departure from a distributive law." The question of this note is: *which categorical invariant is k (equivalently, the weld's non-associativity)?*

2. The method: name the map, then decide

We adopt one discipline, learned the hard way across a month of cross-domain bridge-hopes that died on inspection.

Remark 2.1 (Naming the map buys decidability, not survival). A candidate obstruction home ranked highly because its *vocabulary* matches ("cocycle", "splitting", "Yang–Baxter") has earned nothing until a *structure map* is exhibited between the *specific* objects at hand — not the words. "This sounds like it should connect" is a prior of roughly zero. The payoff of naming a candidate map is not that it survives; it is that the question becomes *decidable*. Our record on resonance-ranked leads is $0/n$; our record on *named*-map leads is: all decided, crisply, one way or the other.

Accordingly we probe three obstruction theories with three named maps. Each verdict below is one of three kinds, and we keep them strictly separate:

- **proved** — a registered result of the program (items (1)–(3));
- **evidence** — a literature-sweep conclusion, *not* a theorem (the handedness-originality claim, §7);
- **open** — an undecided lead (the fourth receptacle, §8).

3. The three decided named maps

3.1. (1) weld $\mapsto$ regular bicomodule — EXISTS (proved).

Theorem 3.1 (Spivak–Garner–Fairbanks, "Functorial Aggregation", arXiv:2111.10968, Prop. 3.9(5) + Thm. 4.28)**.** *For polynomial comonoids (equivalently small categories) c and d, there is an equivalence*

$$\mathrm{Comod}(\mathbf{Poly})(c,d) \;\simeq\; \mathrm{PRA}(c\text{-}\mathbf{Set},\, d\text{-}\mathbf{Set})$$

between (c,d)-bicomodules in **Poly** *and parametric right adjoint functors c-**Set** $\to$ d-**Set**.*

Proposition 3.2 (weld is a regular bicomodule; registry `weld-is-regular-bicomodule`, **proved**)**.** *The weld $B = \Delta S \triangleleft \Delta T$ is the* regular *$(\Delta T, \Delta S)$-bicomodule. Under Theorem 3.1 it represents the prafunctor*

$$\bar{F}_B(V) \;=\; (T \times V)^S \;=\; T^S \cdot V^S,$$

whose slice (aggregation) factorization keeps the power $(-)^S$ as connected-limit content and aggregates over the branch-map set T^S by a coproduct Σ.

This is the *positive* corner of the triangulation: the structure really is there. The result is peer-reviewed within the program (Rick's referee pass, rounds 1 and 2) and carries a Lean-verified leaf checking the five regular-bicomodule coaction axioms. We record it so that the reader sees the method is not merely destructive: when a named map exists, it is found and certified.

Remark 3.3 (The bicomodule is associative; keep it separate from the defect). Crucially, the bicomodule/PRA picture of §3.2 lives in the *associative* regime. The weld, *once its Zappa–Szép data make it well defined*, is a genuine comodule, and the cohomological class $[\omega] \in H^2$ attached to it (elsewhere in the program) is a *splitting* obstruction of an *already- associative* weld, not a measure of non-associativity. The non-associativity we chase here is the Kleisli-side sesquilaw defect — the *departure* from that associative regime. This separation is the hinge of the whole note.

3.2. **(2) $k \mapsto$ 2-cocycle — REFUTED, no receptacle.**

Claim 3.4 (registry: refuted; cf. `h2-weld-class-is-splitting-not-associativity`). The renormalization idempotent $k = n\,;m$ is *not* a 2-cocycle. The refutation is not that the putative class vanishes; it is that the *receptacle does not exist.*

Why the receptacle is empty. A 2-cocycle condition, in every incarnation (group cohomology, the Sweedler/Hopf cocycle, the obstruction cocycle of an extension), *is* an associativity equation: the cocycle identity is precisely the statement that two bracketings of a triple agree. Thus a 2-cocycle presupposes an associative multiplication whose coherence it then measures. A genuinely non-associative magmad has no associative multiplication to begin with, hence no cocycle at all — there is nothing for a class to be a class *of.* One cannot carry the non-associativity as a cohomology class because the apparatus that would carry it is *defined by* associativity. (Confirmed independently by Román's magmads: associativity- failure is a *magma* phenomenon, not a cocycle phenomenon.) □

This is a *well-posed negative of the first kind*: the map was named, checked, and the home turned out to be empty.

3.3. **(3) $k \mapsto$ Karoubi/Cauchy splitting obstruction — REFUTED, vacuous receptacle.**

Here the receptacle *exists*, and that makes the negative subtler and, we think, more instructive.

Lemma 3.5 (k lives in an associative hom-set and is idempotent)**.** *$n\colon DM \Rightarrow MD$ and $m\colon MD \Rightarrow DM$ are morphisms in the functor category $[\mathbf{Set},\mathbf{Set}]$, whose composition (vertical composition of natural transformations) is strictly associative. Hence $k = n\,;m \in \mathrm{End}_{[\mathbf{Set},\mathbf{Set}]}(DM)$, and "is k idempotent?" and "does k split?" are both well posed. Moreover k is idempotent: from the* section identity $m\,;n = \mathrm{id}_{MD}$ *(a normalized distribution, included as a subdistribution, already has unit mass, so renormalizing is the identity),*

$$k\,;k \;=\; (n\,;m)\,;(n\,;m) \;=\; n\,;(m\,;n)\,;m \;=\; n\,;\mathrm{id}_{MD}\,;m \;=\; n\,;m \;=\; k.$$

Theorem 3.6 (sesquilaw k is a split idempotent; registry `sesquilaw-magmoid-karoubi`, **proved**)**.** *$k = n\,;m$ is a* split *idempotent. Its splitting is*

$$DM \xrightarrow{\;n\;} MD \xrightarrow{\;m\;} DM, \qquad m\,;n = \mathrm{id}_{MD}, \qquad n\,;m = k,$$

with splitting object the magmad carrier MD itself, retraction n and section m. Consequently:

1. *the Karoubi/Cauchy (idempotent) completion is a* no-op *on k — k already lies in the image of $[\mathbf{Set},\mathbf{Set}] \hookrightarrow \mathrm{Kar}([\mathbf{Set},\mathbf{Set}])$, so there is no "missing splitting";*
2. *there is* no splitting obstruction *to carry the non-associativity: the feared failure (that k does not split) is empty.*

Proof. A split idempotent is exactly a factorization $e = s\,;r$ with $r\,;s = \mathrm{id}$ through the image; Lemma 3.5 exhibits $r = n$, $s = m$, with $m\,;n = \mathrm{id}_{MD}$ and $n\,;m = k$. The splitting object MD is already present in $[\mathbf{Set},\mathbf{Set}]$, so idempotent completion adds nothing for k. Both consequences follow. □

What makes (3) decisive is not the single computation but the meta-lemma it instantiates.

Lemma 3.7 (idempotent completion preserves associativity)**.** *For any category $\mathcal{D}$, the Karoubi/Cauchy completion* $\mathrm{Kar}(\mathcal{D})$ *is a* category*: its composition is associative. Idempotent completion is an idempotent monad on* **Cat** *landing in Cauchy-complete categories; it preserves and reflects associativity (Borceux I, §6.5; nLab, "idempotent completion").*

Corollary 3.8 (splitting can never measure non-associativity)**.** *Splitting idempotents in an associative $\mathcal{D}$ yields only an* associative *category. Hence no "split some idempotents in an associative core" construction can* produce *a non-associative composition; dually, the failure of a composition to be associative can never be* identified with *the failure of an idempotent to split, because the latter is repaired by a construction (Lemma 3.7) that cannot repair the former. The two defects live in orthogonal parts of the categorical universe.*

Corollary 3.8 is why the candidate map (3) was doomed *before* any computation: even had k failed to split in some ambient $\mathcal{D}$, completing to repair it would land in an associative category, so the repaired object could not witness the magmoid's genuine non-associativity. This is a *well-posed negative of the second kind*: the home exists but carries no obstruction content.

4. The Karoubi vacuity is generic, not weld-specific

Two independent sources, from problems with no connection to the weld, confirm that item (3) is a general feature of the weak-distributive-law family rather than an artifact of our particular k.

- **Böhm–Lack–Street, "Idempotent Splittings, Colimit Completion, and Weak Aspects of the Theory of Monads" (arXiv:1102.4931, 2011).** Weak monad notions — e.g. *demimonads*, which drop the unit law but keep associativity $\beta \cdot T\beta = \beta \cdot \mu$ — reduce systematically to *strict* structure inside the Cauchy-completed $Q_*\mathcal{K}$. That is an independent, pre-weld proof that Cauchy/Karoubi completion *absorbs* one-sided weakenings of the monad laws rather than *detecting* them: exactly Corollary 3.8, reached from a wholly different motivating problem.
- **Fritz–Gonda–Lorenzin–Perrone–Stein (arXiv:2308.00651, 2023).** Every idempotent Markov kernel between standard Borel spaces splits: idempotent-splitting is the *generic* case in a well-behaved category. So $k = n\,;m$ splitting was never going to be a surprise; the vacuity of (3) is the expected case, not a coincidence.

Together these upgrade (3) from "true for our k" to "structurally true for the family": the Karoubi receptacle is vacuous wherever this kind of one-sided-weak structure appears.

5. The positive replacement: one-sided absorptivity (handedness)

If the defect is neither a cocycle nor a splitting obstruction, what is it? The honest answer is concrete and non-cohomological.

Theorem 5.1 (non-associativity = absorptivity asymmetry of the split idempotent)**.** *Model* **Norm** *at the hom level over the associative Kleisli category $\mathcal{C} = \mathrm{Kl}(DM)$ of the honest subdistribution monad DM. On each hom-set put the split idempotent $e = (-)\,;k$, with section $s = (-)\,;m$ and retraction $r = (-);n$, so $r\,;s = \mathrm{id}$ and $s\,;r = (-);k$. Magmoid composition is $f \# g = r\big(s(f)\,;_{\mathcal{C}}\, s(g)\big)$. Then:*

(1) *k is* left*-absorptive: $k\,;n = n\,;m\,;n = n\,;\mathrm{id} = n$, and more strongly a k inserted on a* left *factor is undone by the final normalization, so*

$$(f \# g) \# h \;=\; (a\,;b\,;c)^{\circ} \quad \textit{(normalize once, at the very end),}$$

writing $a = s(f)$, etc.

(2) *k is* not *right-absorptive: a k inserted on a* right *factor rescales the rows of $b\,;c$ by non-uniform scalars $1/\mathrm{mass}_y$, reweighting the mixture and surviving the final normalization, so in general*

$$f \,\#\, (g \,\#\, h) \;=\; \big(a\,;[(b\,;c)\,;k]\big)^{\circ} \;\neq\; (a\,;b\,;c)^{\circ}.$$

Hence $(f \,\#\, g) \,\#\, h = f \,\#\, (g \,\#\, h)$ for all triples iff *k is* also *right-absorptive — which it is not. The non-associativity is precisely the* absorptivity asymmetry *of k.*

The claim "left-absorptive" is arXiv:2510.01131's own Lemma 3.5; the hom-level model reproduces that paper's Proposition 4.9 purity cases exactly: in a finite $\perp$-absorbing model (5000 trials per case) the generic associativity defect is $\sim 7.7 \times 10^{-3}$, while each of the three purity conditions — f S-pure, g S-pure, h T-pure — forces the defect to $\sim 2.2 \times 10^{-16}$, with the opposite purities failing as predicted. The left-bracket equals normalize-at-end to 2×10^{-16}; the right-bracket does not ($\max 8 \times 10^{-2}$).

Remark 5.2 (handedness is a different kind of axis). One-sided absorptivity is not a cohomological class and not a splitting datum. It is a *handedness* axis: a property of *which side* of a composite a distinguished map is invisible on. The defect "lives" there, orthogonal to the two theories that cannot see it.

6. The cross-theme punchline

The two refutations are the *same fact read twice*.

- **Cohomology side** (§3.4): the cocycle *condition equals* associativity, so non-associativity $\Rightarrow$ no cocycle to carry a class. (H^2, when it does appear in the program, is a *splitting* obstruction of an *already-associative* weld.)
- **Karoubi side** (§3.8): idempotent completion *preserves* associativity, so splitting can never *measure* non-associativity.

Two *independent* obstruction theories — group-cohomological (H^2/cocycle) and Cauchy/Karoubi (idempotent-splitting) — are therefore both orthogonal to associativity, and they say so for the *same* structural reason. The content of this note is that convergent negative: **associativity-failure and splitting-obstruction are orthogonal axes**, confirmed from both directions. After a month of cohomological bridge-hopes dying, the weld's non-associativity is measurable by *no* standard obstruction apparatus; the surviving handle is the humble, concrete handedness of k.

7. Originality of the handedness invariant (evidence, not theorem)

We now state a claim of a different epistemic status — an *evidence*-grade conclusion from a literature sweep, deliberately *not* promoted to a theorem.

Claim 7.1 (evidence-grade; literature sweep, 2026-10-22). A dedicated four-agent sweep for a prior categorical home of "one-sided absorptivity of a single distinguished map" crossed with idempotent-completion or associator-laxity theory found *none*. The two genuine precedents are algebra/ideal-flavored, not categorical containers for a single structure map:

- **near-semirings**: a one-sided distributive law together with a one-sided absorbing zero ($0 \cdot a = 0$ required, $a \cdot 0 = 0$ not), e.g. transformation near-semirings $M(\Gamma)$. The handedness-of-absorption $\leftrightarrow$ handedness-of-distributivity link is the right shape, but it is classical algebra, not category theory.
- **Grandis, "Categories with Negation" (arXiv:1709.03186)**: left/right- absorbing morphism *classes* (closed under pre-/post-composition) — an *ideal*, not a single map.

Remark 7.2 (honest scope). This is a negative literature result, not a proof of non-existence. It says: after a focused search, no prior home was found that crosses idempotent-completion/associator-laxity with one-sided absorptivity of a *single* map; the nearest matches are structurally one level off

(an ideal of morphisms, or an algebraic one-sided zero, rather than a single distinguished structure map). We report it as the current state of evidence, and we would retract it the moment a genuine precedent surfaced.

8. Open: a candidate fourth receptacle

The one live technical (non-vocabulary) lead for an intrinsic home of handedness is *lax distributive laws by inequality*.

- **Schoen, "Lax Liftings and Lax Distributive Laws" (arXiv:2307.16552, ACT 2022):** a lax distributive law $\lambda_X \colon FP(X) \to PF(X)$ is defined by a one-directional naturality *inequality*; directionality is baked in by design, not by resonance. This is the closest technical precedent — genuine one-sidedness in the definition — but its shape is relational/coalgebraic (**Set** $\to$ **Rel**, bisimulation), a *different kind* of one-sidedness than composition-operation absorptivity. Structurally adjacent, not a match.
- **Tholen, "Lax Distributive Laws for Topology I/II" (arXiv:1603.06251, 1605.04489):** the same lax-inequality family; a retry target (PDF extraction failed on the first pass).

We flag this as *open*: whether handedness has a fourth, intrinsic categorical home — either by a genuine map into the lax-distributive-law framework, or by a categorical oidification of the near-semiring one-sided zero — remains undecided. Until a structure map is named and checked (§2), it is a watch item, not a result.

9. Grant relevance

The program's obstruction-theory narrative gains a sharpened, honestly-bounded unit: for the central compositional tool (the Zappa–Szép weld of directed containers), we now know precisely *what the non-associativity is not* — not a cocycle, not a Karoubi-splitting obstruction, and (by two independent confirmations) not accidentally so — and we have a positive handle for *what it is*: a one-sided absorptivity (handedness) datum with no prior categorical home. "Handedness datum, not characteristic class" is a crisp, defensible claim about where compositional non-associativity lives, and it is exactly the kind of negative-plus-positive result that keeps the Impact narrative ("obstruction = cohomology") honest about which obstructions are actually ours.

Kodamai